

第3次作业：MPI编程实验

- 实验内容
 - 用MPI编程实现矩阵并行乘法程序
 - 矩阵大小 $\geq 8000 \times 8000$ ，矩阵元素为双精度浮点数 (约需要480MB内存)
 - 编程语言：C/C++
 - 比较并行与串行程序的加速比
 - 注意：
 - 统计计算时间时，应排除准备数据的时间(如果有I/O)
 - 并行程序的性能可扩展性：进程个数可变、局部性、...
- 作业提交
 - 实验报告：实验内容与方法、实验过程、实验结果与分析
 - 源代码及可执行程序压缩包
- 提交方式：<https://spoc.buaa.edu.cn>
- 提交截止日期：5月28日前

4×4 矩阵乘法

$$\begin{bmatrix} a_{0,0} & a_{0,1} & a_{0,2} & a_{0,3} \\ a_{1,0} & a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,0} & a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,0} & a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \times \begin{bmatrix} b_{0,0} & b_{0,1} & b_{0,2} & b_{0,3} \\ b_{1,0} & b_{1,1} & b_{1,2} & b_{1,3} \\ b_{2,0} & b_{2,1} & b_{2,2} & b_{2,3} \\ b_{3,0} & b_{3,1} & b_{3,2} & b_{3,3} \end{bmatrix} = \begin{bmatrix} c_{0,0} & c_{0,1} & c_{0,2} & c_{0,3} \\ c_{1,0} & c_{1,1} & c_{1,2} & c_{1,3} \\ c_{2,0} & c_{2,1} & c_{2,2} & c_{2,3} \\ c_{3,0} & c_{3,1} & c_{3,2} & c_{3,3} \end{bmatrix}$$

$$A_{0,0} = \begin{bmatrix} a_{0,0} & a_{0,1} \\ a_{1,0} & a_{1,1} \end{bmatrix} \quad A_{0,1} = \begin{bmatrix} a_{0,2} & a_{0,3} \\ a_{1,2} & a_{1,3} \end{bmatrix} \quad A_{1,0} = \begin{bmatrix} a_{2,0} & a_{2,1} \\ a_{3,0} & a_{3,1} \end{bmatrix} \quad A_{1,1} = \begin{bmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{bmatrix}$$

$$B_{0,0} = \begin{bmatrix} b_{0,0} & b_{0,1} \\ b_{1,0} & b_{1,1} \end{bmatrix} \quad B_{0,1} = \begin{bmatrix} b_{0,2} & b_{0,3} \\ b_{1,2} & b_{1,3} \end{bmatrix} \quad B_{1,0} = \begin{bmatrix} b_{2,0} & b_{2,1} \\ b_{3,0} & b_{3,1} \end{bmatrix} \quad B_{1,1} = \begin{bmatrix} b_{2,2} & b_{2,3} \\ b_{3,2} & b_{3,3} \end{bmatrix}$$

$$C_{0,0} = \begin{bmatrix} c_{0,0} & c_{0,1} \\ c_{1,0} & c_{1,1} \end{bmatrix} \quad C_{0,1} = \begin{bmatrix} c_{0,2} & c_{0,3} \\ c_{1,2} & c_{1,3} \end{bmatrix} \quad C_{1,0} = \begin{bmatrix} c_{2,0} & c_{2,1} \\ c_{3,0} & c_{3,1} \end{bmatrix} \quad C_{1,1} = \begin{bmatrix} c_{2,2} & c_{2,3} \\ c_{3,2} & c_{3,3} \end{bmatrix}.$$

$$\begin{bmatrix} A_{0,0} & A_{0,1} \\ A_{1,0} & A_{1,1} \end{bmatrix} \times \begin{bmatrix} B_{0,0} & B_{0,1} \\ B_{1,0} & B_{1,1} \end{bmatrix} = \begin{bmatrix} C_{0,0} & C_{0,1} \\ C_{1,0} & C_{1,1} \end{bmatrix}$$

$$C_{0,0} = A_{0,0} \times B_{0,0} + A_{0,1} \times B_{1,0}$$

$$C_{0,1} = A_{0,0} \times B_{0,1} + A_{0,1} \times B_{1,1}$$

$$C_{1,0} = A_{1,0} \times B_{0,0} + A_{1,1} \times B_{1,0}$$

$$C_{1,1} = A_{1,0} \times B_{0,1} + A_{1,1} \times B_{1,1}$$